Why  ?

Outline

I. Who am I?
II. My R journey
III. What/why 
IV. What/why 

Who am I?

Katherine Muller, MS PhD

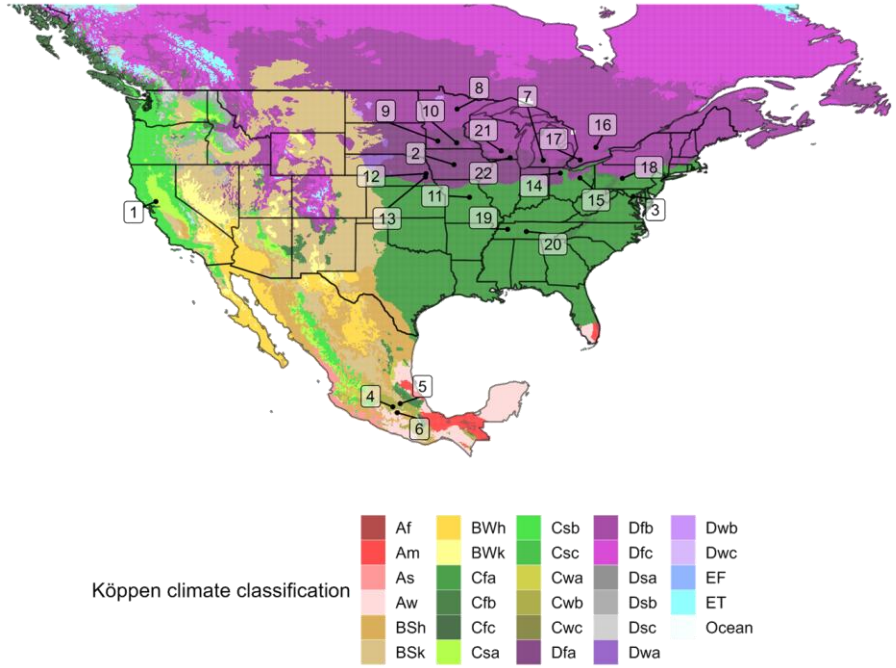
Agroecologist

Applied Statistician

Data Scientist

Self-taught!

D.R.I.V.E.S.

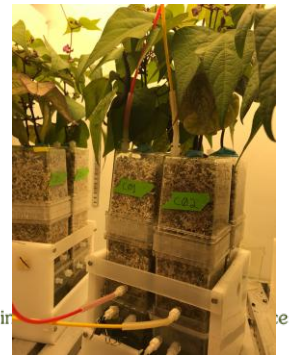


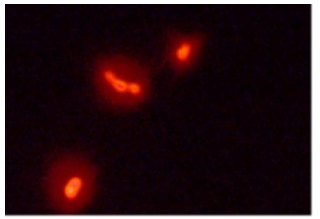
My training

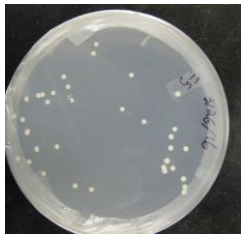
- Dual Bachelor's in Biology and Music
- Master's and PhD in plant evolutionary ecology
- Postdoc in legume cover crop agroecology









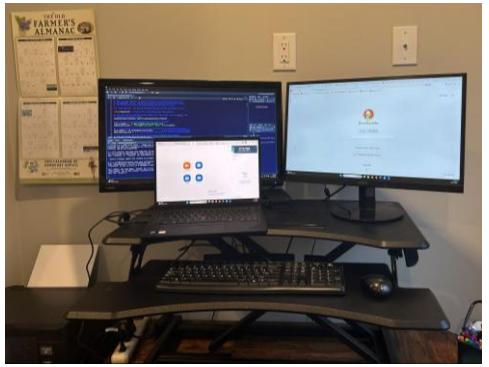


My research approaches

- Molecular biology (just a little)
- Field, lab, and greenhouse experiments with plants and microbes
- Bayesian statistics
- Statistical and theoretical modeling
- Complex data management







My R Journey

2012



Ecological Entomology

Ecological Entomology (2016), **41**, 51–60 DOI: 10.1111/een.12257

Echinacea angustifolia and its specialist ant-tended aphid: a multi-year study of manipulated and naturally-occurring aphid infestation

KATHERINE E. MULLER[1] and STUART WAGENIUS[2] [1]Program in Plant Biology and Conservation, Northwestern University, Evanston, Illinois, U.S.A. and [2]Chicago Botanic Garden, Glencoe, Illinois, U.S.A.

ROYAL SOCIETY OPEN SCIENCE

rsos.royalsocietypublishing.org

Research

Cite this article: Muller KE, Denison RF. 2018 Resource acquisition and allocation traits in symbiotic rhizobia with implications for life-history outside of legume hosts. *R. Soc. open sci.* **5**: 181124. http://dx.doi.org/10.1098/rsos.181124

Resource acquisition and allocation traits in symbiotic rhizobia with implications for life-history outside of legume hosts

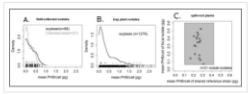
Katherine E. Muller[1] and R. Ford Denison[2]

[1]Graduate Program in Plant and Microbial Biology, University of Minnesota, St Paul, MN 55108, USA
[2]Department of Ecology Evolution and Behavior, University of Minnesota, St Paul, MN, USA

KEM, 0000-0001-9144-3867

Data and analysis scripts for the paper: Resource acquisition and location traits in symbiotic rhizobia with implications for life-histo outside of legume hosts

Muller, Katherine E; Denison, R Ford (2018-10-15)



Title

Data and analysis scripts for the paper: Resource acquisition and allocation traits in symbi zobia with implications for life-history outside of legume hosts

Published Date

2018-10-15

Persistent link to this item

https://doi.org/10.13020/D69H6N
https://hdl.handle.net/11299/200613

Authors

Muller, Katherine E
Denison, R Ford

Crop Science

CSSA Crop Science Society of America

ORIGINAL RESEARCH ARTICLE

Estimating agronomically relevant symbiotic nitrogen fixation in green manure breeding programs

Katherine E. Muller, Joseph Guinness, Matthew Hecking, Laurie E. Drinkwater

First published: 27 March 2021 | https://doi.org/10.1002/csc2.20517 | Citations: 6

Crop Science

CSSA Crop Science Society of America

ORIGINAL ARTICLE | Open Access

Genetic and environmental drivers of legume cover crop performance: Hairy vetch

Lisa Kissing Kucek, Katherine Muller, Lais Bastos Martins, Virginia M. Moore, Chris Reberg-Horton, Steven B. Mirsky, John Englert, Laurie E. Drinkwater, Joel Douglas, Sarah S. Eagen ... See all authors

Agronomy Journal

ASA American Society of Agronomy

ORIGINAL ARTICLE | Open Access

Prolonged soybean absence in the field selects for rhizobia that accumulate more polyhydroxybutyrate during symbiosis

Katherine E. Muller, R. Ford Denison

First published: 16 May 2024 | https://doi.org/10.1002/agj2.21578 | Citations: 2

ARTICLE · Volume 7, Issue 9, P1638-1654, September 20, 2024 · *Open Access* Download Full Issue

Rotational complexity increases cropping system output under poorer growing conditions

K. Ann Bybee-Finley [1,2,20,22] · Katherine Muller [1,20] · Kathryn E. White [1,3] · ... · Nele Verhulst [4] · Brook Wilke [18] · Timothy M. Bowles [19,21] ... Show more

Affiliations & Notes · Article Info

Data from: Estimating agronomically relevant symbiotic N fixa tion in green manure breeding programs



Author(s)

Muller, Katherine
Guinness, Joseph
Hecking, Matthew
Drinkwater, Laurie

Files

MULLER_SNFmethodsPaper_Readme.txt (9.77 KB)

Code and limited data for: Rotational complexity increases cropping system output under poorer growing conditi

Cite | Download all (1.87 MB) | Share | Embed | + Collect

dataset posted on 2024-06-26, 16:43 authored by Katherine Muller, Ann Bybee-Finley, Harry H. Schomberg, Michel Cavigelli, Kathryn E. White, E. (Eunjin) Han, Tim Bowles, Frederi Viens

USAGE METRICS

1 111 | 11 666 | 0

Data and analysis scripts for: Prolonged soybean absence in the field selects for rhizobia that accumulate more polyhydroxybutyrate during symbiosis

Cite | Download all (260.06 MB) | Share | Embed | + Collect

README License MIT license

drivesR

The goal of drivesR is to facilitate data management and analysis for members of the Diverse Rotations Improve Valuable Ecosystems Services (DRIVES) Project and anyone interested in using the publicly-available portions of the DRIVES database.

The DRIVES database contains yield and weather data from 21 long-term agricultural field experiments with a crop rotation component. You can find out more about our research and participating sites at our website, drives-network.org/. Metadata and a full data inventory can be found at the DRIVES database repository on Ag Data Commons: https://hdl.handle.net/10779/USDA.ADC.28654694.

As of May 2025, access to DRIVES database requires an API key. Project members and collaborators can request an API key from the DRIVES data manager. Other users must complete our data use survey and agree to our data use terms. Outside users will have read access to a subset of the data excluding sites and site years embargoed from public access.

Installation

You can install the development version of drivesR from GitHub with:

```
# install.packages("devtools")
devtools::install_github("DRIVES-Project/drivesR")
```

Example

This vignette provides detailed instructions and examples for accessing and harmonizing DRIVES data tables.

D.R.I.V.E.S.

QUICK FACTS

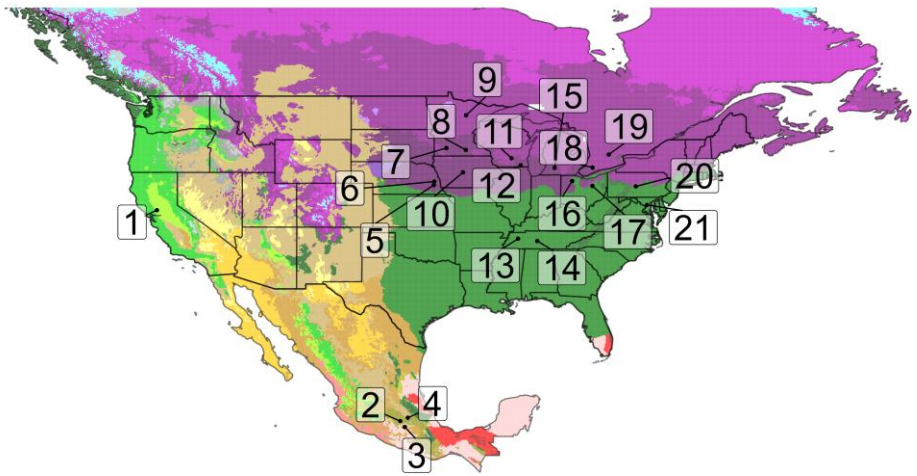
22 long-term field experiments

495 site-years of data

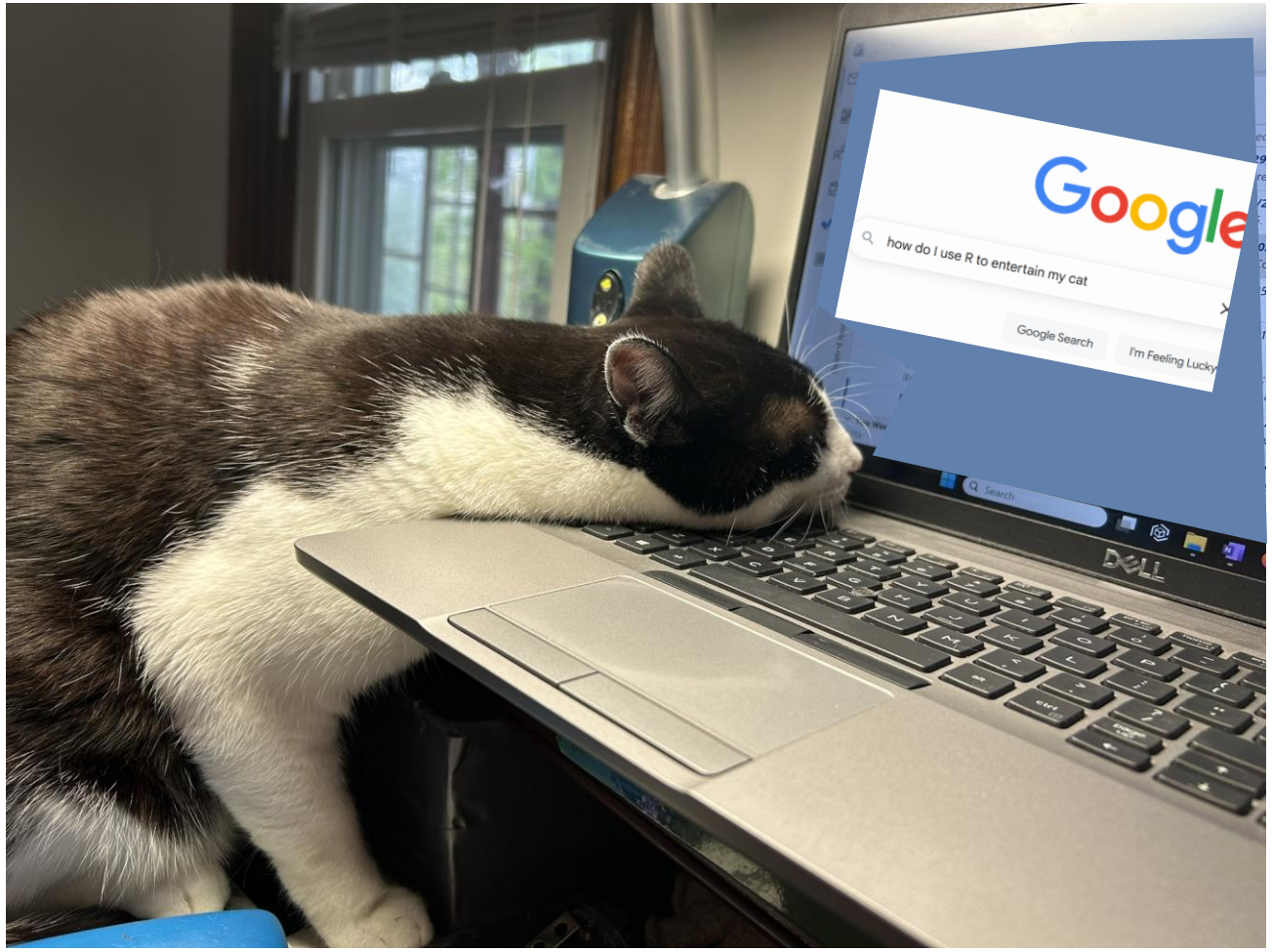
87,300 crop yield observations

311,600 daily weather observations

18 sites with long-term soils data



Still learning



What is R?



- A computer programming language
- Expansive set of tools for data science and statistics



Statistics



Machine learning

Why use R?

- Free!
- Open source



Shared and developed by a global community of users

Why use R?

- R can do anything!

https://r-graph-gallery.com/

https://rmarkdown.rstudio.com/gallery.html

Manipulate data

Visualize

Analyze

Simulate

Communicate

Automate



- Like apps for your phone
- Anyone can make one!

‘soilpalettes’ package

README MIT license



soilpalettes

DOI 10.5281/zenodo.3893248 R-CMD-check no status

This package contains color palettes for RStudio, inspired by soil profiles. Palettes are colorblind-friendly and were created using Chroma.js and Coolors.

Source images and color hex values can be found in the `images.md` file in the root directory.

install package

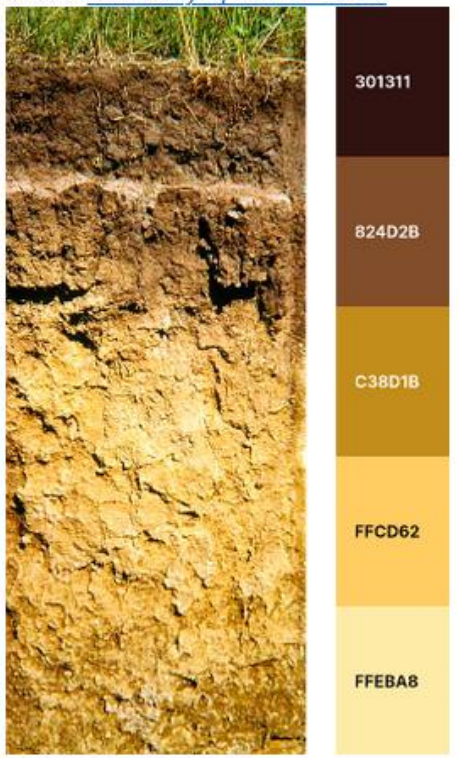
```
# install.packages("devtools")
devtools::install_github("kaizadp/soilpalettes")
```

- ▸ Palettes
- ▸ Examples
- ▸ Acknowledgments and Contributions

alaquod Aeric Alaquod, NC.
source University of Idaho - CALS



natrudoll
source University of Idaho - CALS



‘cowsay’ package

```
> cowsay::say("Isn't this a fun way to proc
rastinate?", by = "cat")

 ______________________________________
< Isn't this a fun way to procrastinate? >
 --------------------------------------
         \
          \

            |\___/|
          ==) ^Y^ (==
            \  ^  /
             )=*=(
            /     \
            |     |
           /| | | |\
           \| | |_|/\
      jgs  //_// ___/
                \_)
```

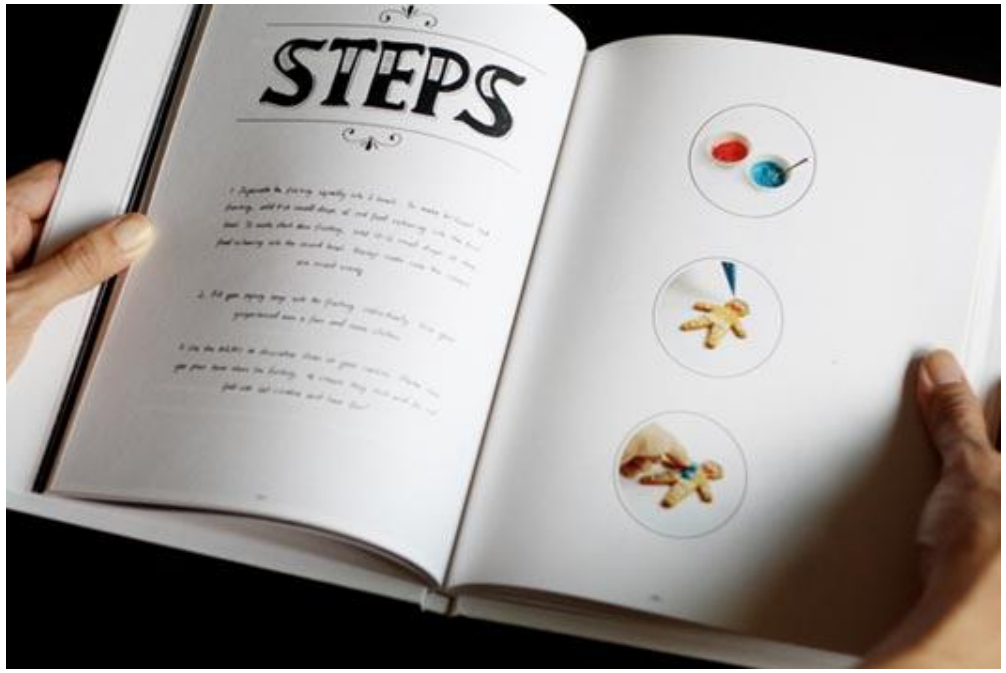
```
> cowsay::say("Sure is!", by = "cow")

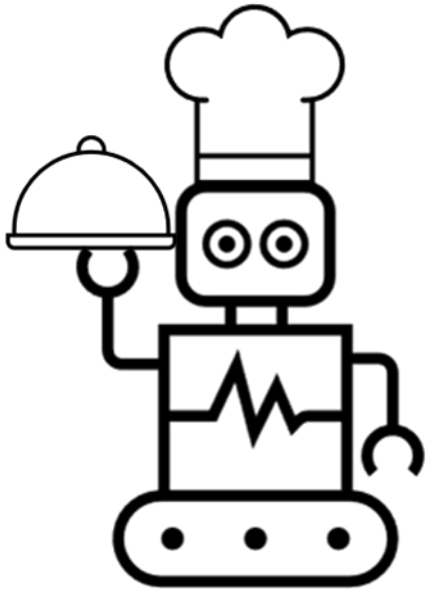
 ___________
< Sure is! >
 -----------
        \
         \

          ^__^
          (oo)\ ________
          (__)\         )\ /\
               ||------w|
               ||      ||
```

Why use R?

- Reproducibility





Why use R?

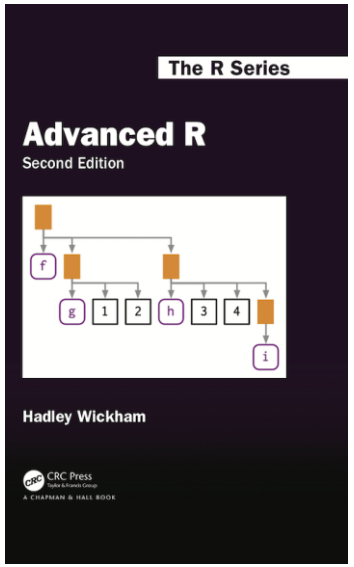
- Extensive R learning community

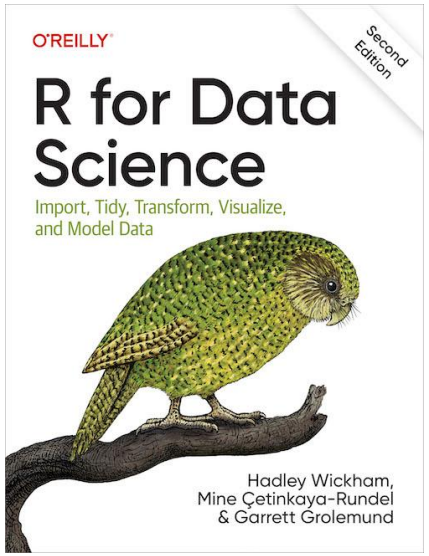
posit

StackExchange

R Studio Education







Free online courses

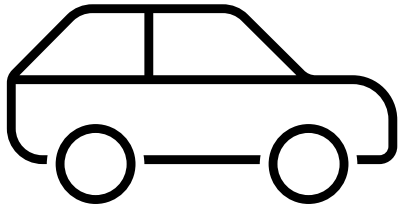
YouTube

What is R studio?

IDE = Integrated Developer Environment

- An environment to work with R







- A set of tools to help your workflow

What is R studio?



RStudio Desktop

- A software application you download and install on your computer
- Uses files on your hard drive
- Open source

Posit Cloud

- Software you run online through your browser (no installation needed)
- Uses files on a cloud server
- Tiered access with user account (free and paid versions)

Why use R studio?



- FREE and open source*
- Makes complicated tasks easier

Packages | Version control | Templates | Integrations

Next: How to install R and Rstudio